

VIGNESH KUMAR EKAMBARAM

TECH LEAD

Full-Stack Architecture | Client Delivery | Team Leadership

Chennai, Tamil Nadu, India | 12+ years in software development | Open to international and on-site assignments

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PROFESSIONAL PROFILE

Tech Lead with 12+ years of experience across full-stack web applications, desktop software, and Android development. Leads a five-member team and coordinates client requirements, sprint planning, releases, and delivery. Combines .NET Core, Angular, React, and AWS experience with machine-learning and AI support initiatives. International experience includes client support in Thailand and a three-month assignment in Japan.

TECHNICAL SKILLS

Languages & Frameworks

C#, TypeScript, JavaScript, Python, SQL, .NET Core, Web API, EF Core, WCF, WPF, Angular, React, Node.js, Express.js, YARP

Cloud & DevOps

AWS EC2, S3, ELB/ALB, CloudSearch, RDS, WAF, Docker, GitLab CI/CD, Firebase, Vercel, IIS, Nginx, Apache, Linux

Data & Distributed Systems

SQL Server, MySQL, SQLite, Oracle, Query Store, Redis, RabbitMQ, AWS SQS, SignalR, WebSockets

AI, ML & Quality

Qdrant, RAG, AI Agents, MCP, HOG, YOLO, scikit-learn, DecisionTreeRegressor, Playwright, SOLID, design patterns

PROFESSIONAL EXPERIENCE

Tech Lead

alfaTKG
2022 - Present

- **Projects:** Quote, JQMS, GSQ, PTE, Viewer, and Ticket Tracker. **Technologies:** Angular, React, .NET Core, Node.js, SQL Server, MySQL, AWS, machine learning, RAG, Qdrant vector database, and LLMs.
- **Team leadership:** Lead a five-member engineering team, assign responsibilities, track progress, and review code to maintain delivery quality.
- **Client and project delivery:** Work directly with clients and support teams to gather requirements, plan development, and coordinate delivery. Manage sprint plans, release schedules, and deliverable reviews.
- **Task management:** Develop Ticket Tracker and coordinate issue tracking, task allocation, and team workloads.
- **On-site client support:** Visit client sites in Thailand to support implementation, resolve issues, and keep project delivery aligned with client expectations.
- **Machine learning:** Train decision-tree regression models on historical data to predict manufacturing process times and quotation costs. Collaborate with the AI team to evaluate deep-learning approaches for model improvement.
- **AI-assisted support:** Support the development of application-specific agents and an agent orchestrator for chat support across multiple applications.
- **RAG and vector retrieval:** Implement retrieval-augmented generation (RAG) workflows by storing and indexing unstructured content in the Qdrant vector database for semantic retrieval.
- **LLM-powered SQL retrieval:** Use large language models (LLMs) to generate SQL queries and retrieve structured data from relational databases.

- **Test automation:** Use Playwright with MCP-enabled workflows to automate testing, reduce manual testing effort and execution time, and improve test quality.

Senior Full-Stack Developer

alfaTKG
2018 - 2022

- **Projects:** GPN Scheduler and Inspection. **Technologies:** Angular, React, .NET Core, Node.js, SQL Server, MySQL, and AWS.
- **Backend and cloud development:** Developed backend APIs and managed AWS EC2, RDS, CloudSearch, Elastic Load Balancing (ELB), and S3 resources.
- **Application design:** Planned Angular components, API controllers, business logic, and SQL table structures to support maintainable application development.
- **Production support:** Investigated and resolved application-server and database issues in production.
- **International client engagement:** Completed a three-month assignment in Japan, coordinated with clients, supported the company booth at the MF-Tokyo manufacturing exhibition, and installed software at three client sites in different locations.
- **Team coordination and deployment:** Coordinated developers and tracked delivery progress. Developed a shared deployment system for use across projects.

Software Development Engineer

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Aug 2015 - 2018

- **alfaDOCK:** Developed Angular UI features and contributed to .NET Core API development, SQL Server queries, and database access.
- **Plugin architecture:** Contributed to a design in which the alfaDOCK .NET Core API acted as the master API and other project APIs operated as plugins. Used Observer/subscriber patterns to communicate notifications and actions from the master API to plugins.
- **Engineering practices:** Applied object-oriented programming and software design principles when developing application features.
- **Socket App:** Developed a WPF desktop application with SQLite to configure different socket connections and collect data from multiple sources at client sites.

Software Engineer

Sirpi Software Pvt. Ltd.
Oct 2013 - Aug 2015

- **SPCAD:** Implemented small application features, fixed defects, and worked on C++ wrapper classes for integration with .NET applications. Reported progress and issues to a senior developer.
- **Grocery Delivery App (Java):** Developed UI features for customer and delivery-agent Android applications, integrated Google Maps, and resolved UI defects.
- **Android game development:** Implemented functions for updating and tracking object positions and contributed to game-loop logic.

EDUCATION

B.E. in Electronics and Communication Engineering | 2009 - 2013
Hidusthan College, Coimbatore, Anna University | GPA: 8.4

Vignesh Kumar Ekambaram | Tech Lead